THE ENVELOPE EXPLORER

User guide

how to read, filter and trace every message the platform emits

At a glance	
Where it runs	http://127.0.0.1:8802
What it shows	every envelope recorded by the lake, live
Read-only	it never writes to the platform or the bus
Retains	the most recent 20,000 envelopes in memory, plus everything on disk
Built for	forensics, compliance and model analysis

This guide is task-oriented. Each section answers one question: how do I find a message, how do I read one, how do I see what led to a decision, and how do I prove what happened.

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1 Before you start

The Explorer is a viewer. It reads the lake that the recorder writes; it does not talk to the exchange and it cannot change anything. If it is empty, the problem is upstream, not here.

1.1 What should be running

Service	Port	Why the Explorer needs it
market data	:8796	the source of every price, depth and candle envelope
recorder	:8801	consumes the bus once and writes the lake the Explorer reads
envelopeapi	:8802	the Explorer itself

Start them in that order. The Explorer will start without the recorder, but it will have nothing to show.

1.2 Opening it

1. Start the core, then the Explorer (scripts/start.sh does both in order).
2. Open http://127.0.0.1:8802 in a browser.
3. The header should show a rising envelope count within a few seconds.

If the header shows 20,000 and stops climbing, the Explorer is fine and the feed has stopped. Check the recorder's /stats before touching the Explorer.

2 The screen at a glance

Four regions, always in the same place.



Figure 1 — the Explorer with an envelope selected: filters on the left, the list in the middle, the envelope's anatomy on the right.

Region	What it is for
Header	counts: envelopes held, bytes, planes, topics. The 'viewing as' role is on the right.
Timeline	a histogram of traffic over the retained window. Drag across it to select a time range.
Left panel	filters: plane, topic, symbol, producer, correlation id, free text.
Middle	the list of envelopes (newest first) or the journey flow diagram.
Right	the anatomy of the selected envelope: meta, payload, references, chain.

2.1 Breadcrumbs, not modes

The middle pane is a contextual surface. With nothing selected it is the live feed. Select an envelope and it recomposes around that envelope: its identity, its verdict if it is a decision, the envelopes it read or the ones that read it, and buttons to the flow and the full trace.

The bar above it is a breadcrumb — FEED / symbol · topic — not a row of equal tabs. You do not choose a mode; the selection decides what is worth showing. The deeper views (flow, trace, why) open from the context's buttons.

The ← button at the left of the crumb bar is back. It is greyed out when there is nothing to go back to, and it is the same action as Escape: one step, restoring the view and the envelope you had selected.

2.2 Resizing: every panel

Three splitters: one between the filters and the list, one between the list and the detail pane, and one under the timeline. Drag any of them; double-click is not needed because the size persists in the browser, so the layout you set is the layout you get next time.

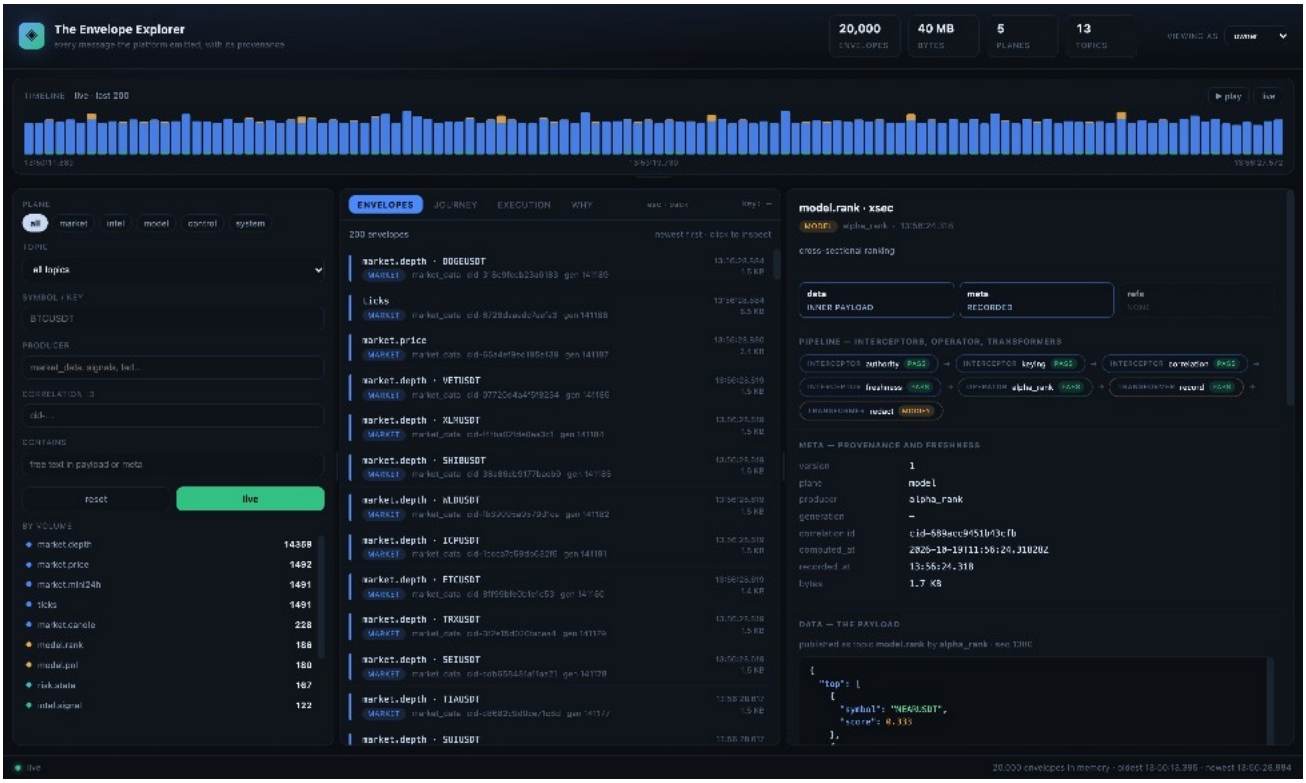


Figure 2 — the filters pane dragged wider, a model decision selected, and the context strip showing the four interceptors that passed before the operator ran.

2.3 What the surface decides

The surface is a set of rules, not a template. Given a selection it asks what kind of thing it is holding and composes accordingly.

The selection	What the surface shows
nothing	the live feed and the timeline
a decision (model.decision)	its verdict: traded or refused, the reason verbatim, the confidence
an outcome (a fill, a ranking)	what it read, by stage, with the age of each input at execution
an input with downstream traffic	what read it next, in time order
an input with no downstream	the chain it belongs to, and a plain statement that nothing has read it yet
a multi-symbol snapshot	an explanation that it carries every symbol, so it has no per-symbol chain

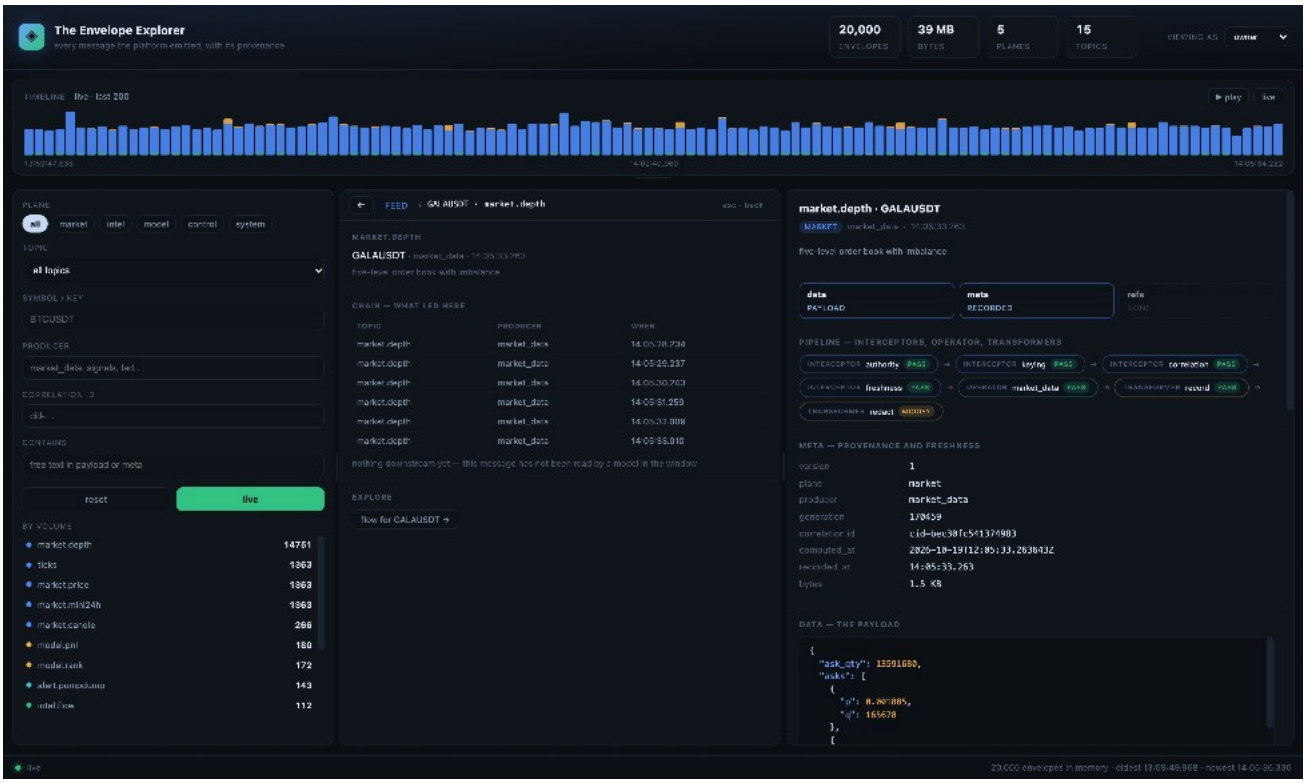


Figure 3 — an input with no downstream: the surface shows the chain that led to it and says plainly that no model has read it in the window.

The rule the surface follows: never show an empty panel. If there is no downstream traffic, show upstream; if there is no chain, say why — a multi-symbol snapshot has none by construction.

2.4 Testing the surface

Six checks that exercise the rules above. Each takes seconds and each has an observable outcome, so a regression shows up as a wrong panel rather than a vague feeling that the UI is off.

#	Do this	Expect
1	open the Explorer with nothing selected	feed plus timeline; no detail, no chain
2	select a market.depth row	chain of that symbol, and the note that nothing downstream yet
3	select a model.decision row (WHY, then the row)	verdict block with traded/refused and the reason
4	select a model.fill row	what it read, with per-stage ages in seconds

5	press ← or Escape	the previous envelope and view come back, not a blank pane
6	drag both splitters, then reload	the layout is exactly as you left it

If a check fails, the cause is usually data rather than layout: the decisions are rare, the in-memory window covers the last 20,000 envelopes, and a symbol with no traffic will legitimately have nothing to show.

3 Finding the message you want

Filters combine: each one narrows the list further. The count above the list tells you how many matched.



Figure 2 — the plane filter set to intel. Every row shows a green 'INTEL' pill and the topic counts on the left change with it.

Filter	Use it when	Example
Plane	you want one class of traffic only	market, intel, model, control, system
Topic	you know the exact kind of message	intel.forecast, model.decision
Symbol / key	you are following one instrument	BTCUSDT
Producer	you suspect one component	market_data, signals, tad
Correlation id	you have an id from a log or a report	cid-9f2a41…
Contains	you remember a value but not the topic	0.0431 or 'liquidation'

Filters match case-insensitively and apply to the symbol, producer, correlation id and free text. The topic list under 'by volume' is also a filter: click a topic to select it.

3.1 When nothing matches

An empty list is information. The header count drops to zero, the list says so plainly, and the timeline above it keeps filling — which tells you the feed is alive and the filters are the reason you see nothing.

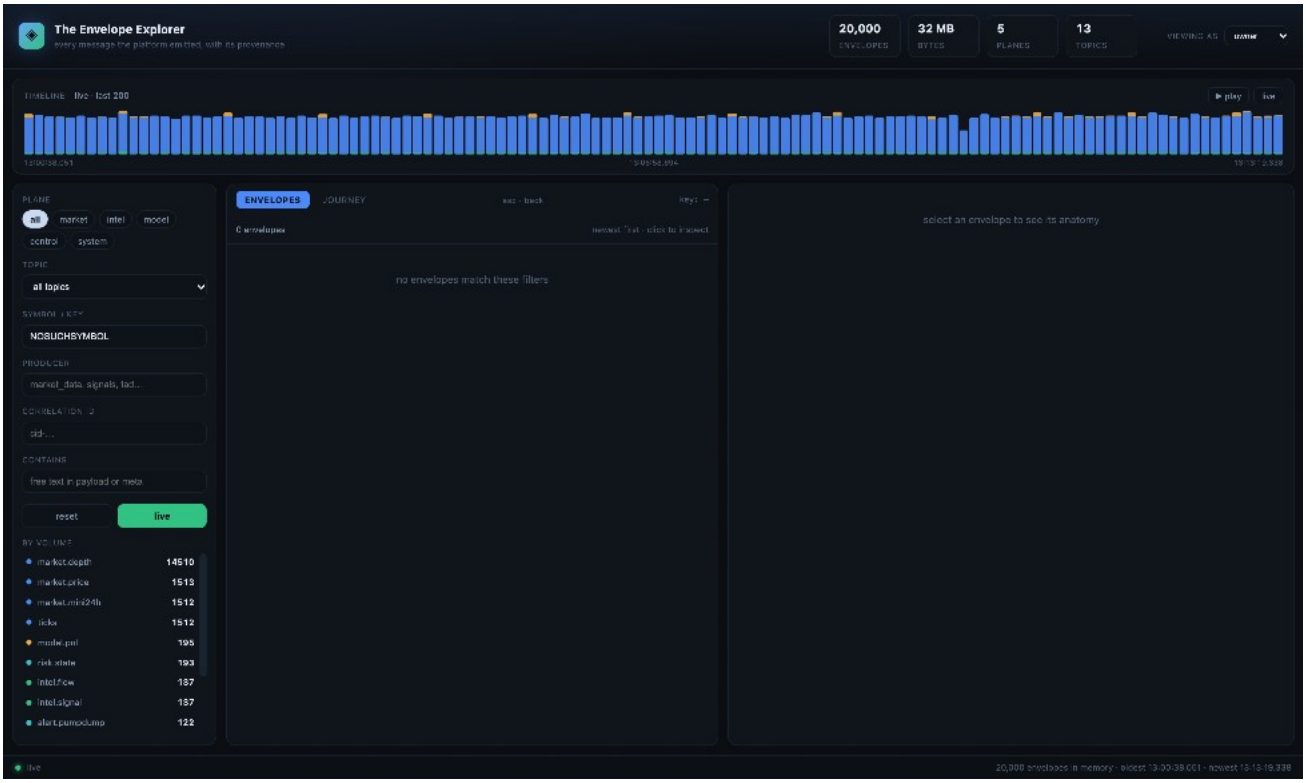


Figure 8 — a symbol filter that matches nothing: 0 envelopes, while the timeline keeps receiving traffic.

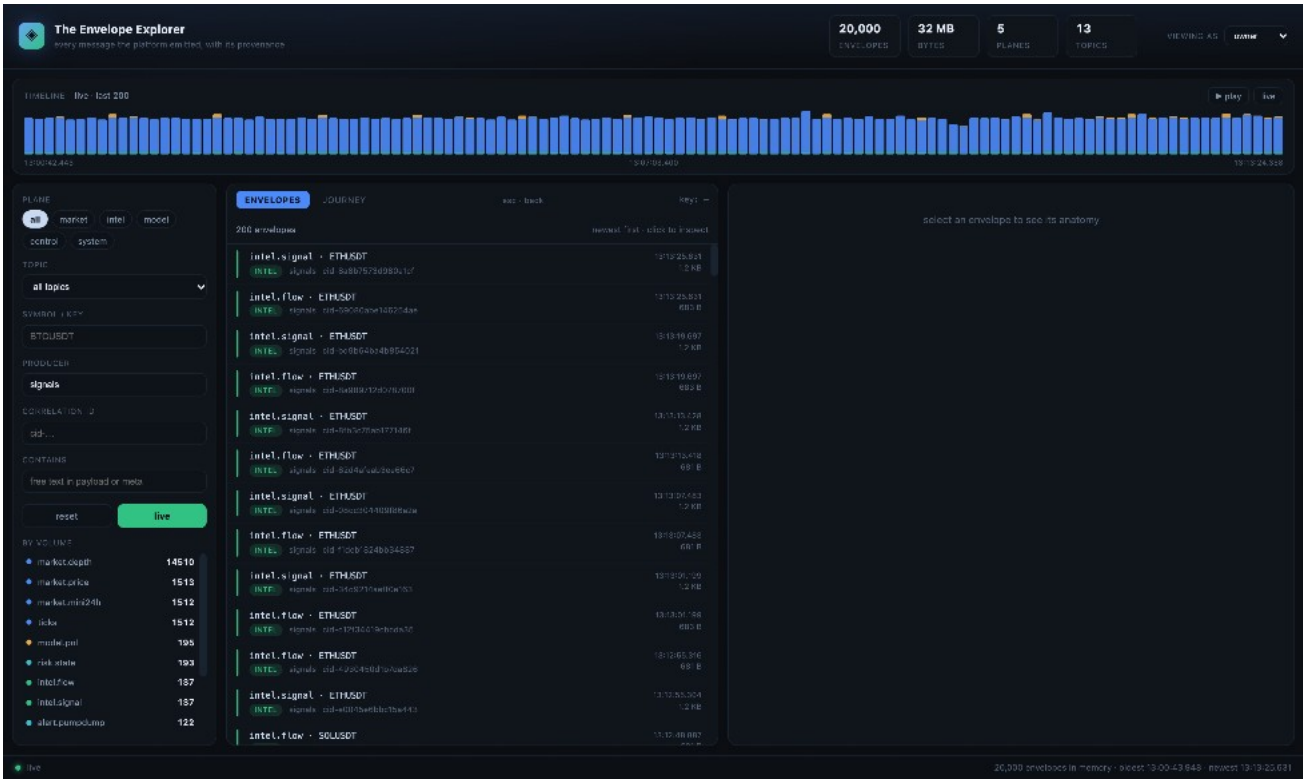


Figure 9 — filtering by producer: every row is a message that component wrote.

3.2 Live versus paused

The green LIVE button means new envelopes are being prepended as they arrive. Set to PAUSED to freeze the list while you read. Selecting a time range on the timeline also pauses it, because you are no longer looking at 'now'.

4 Reading an envelope

An envelope has three members and the right-hand pane shows all three, always: data, meta and references. A strip at the top of the pane tells you at a glance which are present.

4.1 The three members

Member	What it carries	If it is missing
data	the payload for this topic	never — a message with no data is refused
meta	version, plane, producer, generation, correlation id, computed_at, recorded_at, bytes	the pane says 'inferred'
refs	topic, key, generation and correlation id of each input	the pane says 'none' and explains what a reference would contain

References come in two kinds, and the pane always says which: **declared** by the producer, or **derived** by the recorder. The books do not declare their inputs yet, so the recorder attaches, for every model decision, the newest envelope of each input topic for that symbol in the preceding five minutes — and marks each one derived, with its age. A reference is never silently presented as declared when it is inferred.

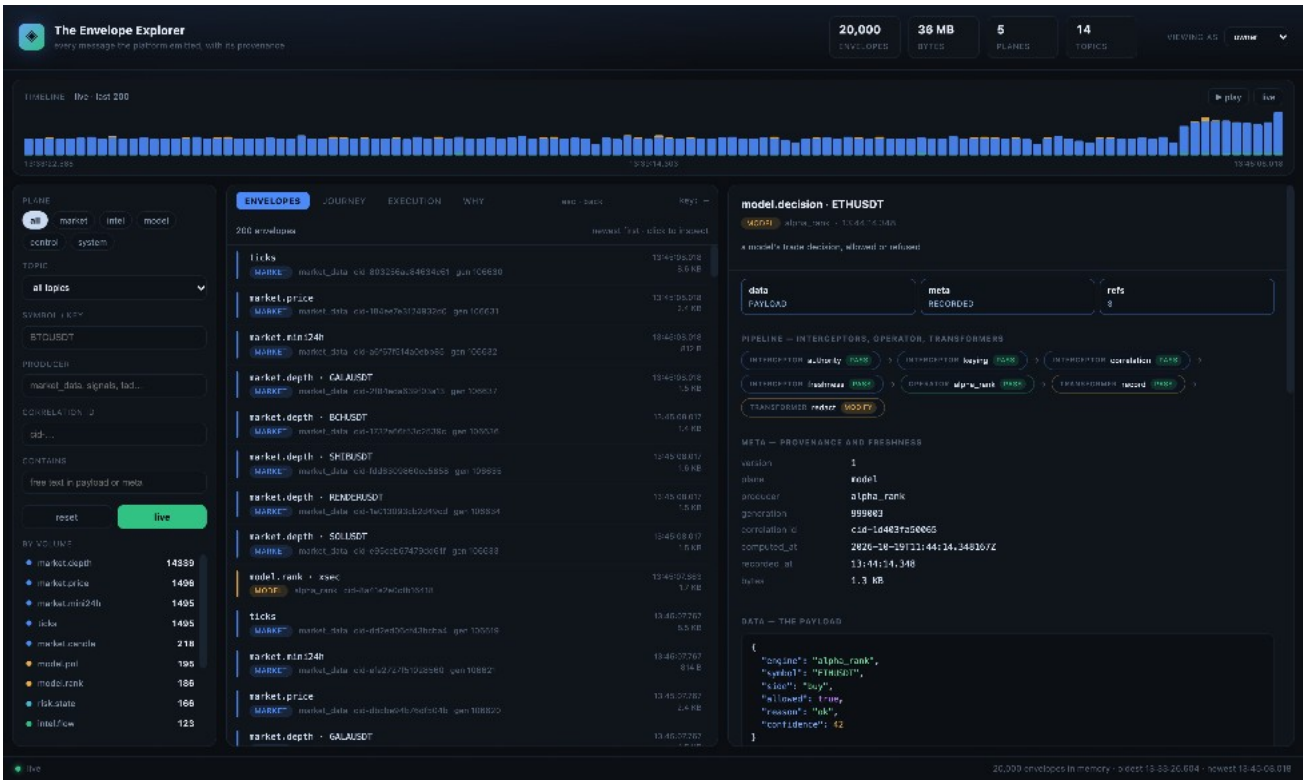


Figure 7 — a decision with three derived references: market.depth at 0.2 s, intel.signal and intel.flow at 1.8 s. The refs member shows 3, and the section names its own provenance.

When a producer publishes its own wrapper (the Python books do: topic, producer, data, version, seq), the pane unwraps it: the Data section shows the inner payload and names the topic and producer it was published under.

4.2 Meta — provenance and freshness

Field	Meaning	Why it matters
version	the envelope contract revision	a reader refuses a revision it does not understand
plane	market, intel, model, control or system	decides priority and how it is handled
producer	the component that wrote it	the first question in any incident
generation	the producer's own counter for that topic	gaps reveal a dropped producer

correlation id	the identity that follows the message	resolves the chain around it
computed_at	when the payload was produced	this is what staleness gates refuse on
recorded_at	when the recorder stored it	the difference is transport and queue time
bytes	size on disk	market data is bytes, and bytes have a budget

4.3 Data — the payload

The payload exactly as it travelled, syntax-coloured. It is never rewritten by the Explorer: what you see is what the recorder stored.



Figure 6 — one decision envelope: meta on top, payload in the middle, chain underneath.

4.4 Pipeline — interceptors, operator, transformer

Every envelope carries the stages it passed through: the interceptors that checked it, the operator that produced it, and the transformers that changed its shape. This is the ATLAS pipeline made visible on the packet itself rather than in a log beside it. A refusal is recorded as an envelope too, so 'why not' is a packet you can open rather than a line you have to find.

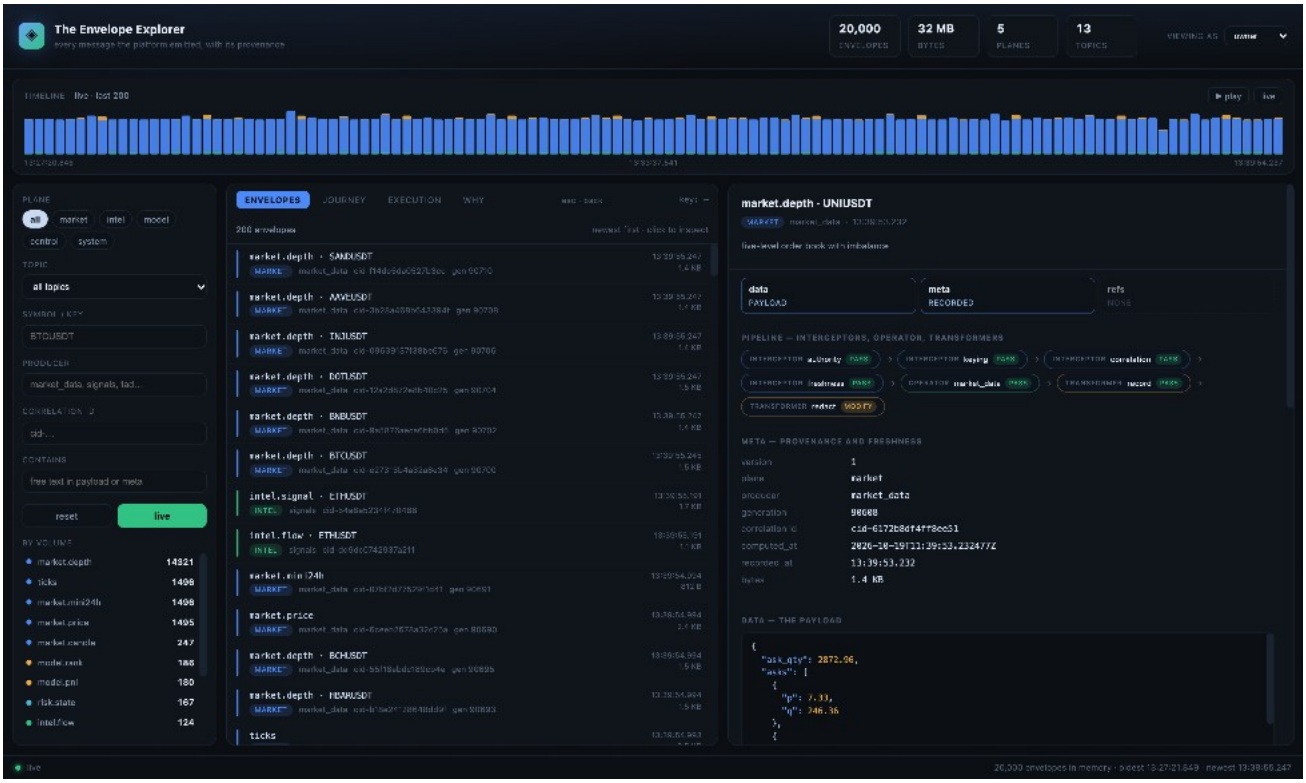


Figure 8 — the pipeline strip: four interceptors, the operator, and two transformers, each with its outcome. The last one is the view-time redaction applied for this role.

Stage	What ran	What the chip means
interceptor	authority, keying, correlation, freshness	a pass means the rule was satisfied; a refusal names the rule and the number
operator	the producer (market_data, signals, a model)	the capability that produced this payload
transformer	record, redact	a change of shape: stored once, or redacted for the viewing role

Outcome colours: green pass, red refuse, amber modify. Hover any chip for its detail — for a refusal that is the exact text the governor recorded, for a transformer it is what changed.

4.5 Chain — what else happened for that symbol

Every envelope recorded for the same symbol, oldest first, with arrows. Click any step to jump to it. This is the quick way to walk a symbol's recent history without leaving the page.

A message with no key (for example market.mini24h) has no chain. That is not a bug: it is a multi-symbol snapshot, so there is no single chain to show.

5 The timeline: browsing history

The histogram above the list is traffic over time, one bar per slice, coloured by plane. It is how you find a moment rather than a message.



Figure 3 — a time range selected by dragging across the bars. Everything outside dims, the header names the window, and the list narrows to it.

1. Drag across the bars to select a window. The list reloads to that window and the live pill turns to paused.
2. Press ▶ play to step the window forward, so the retained period plays back like a recording.
3. Press 'live' to return to the newest 200 envelopes.

Two details worth knowing. The selection survives the histogram's own refresh, so the bars do not jump under the cursor while you are reading. And a gap in the histogram is a real gap: the Explorer never interpolates traffic that was not recorded.

6 The journey: a flow diagram from a point

This is the view to use when the question is 'how did this happen', rather than 'what is this message'.

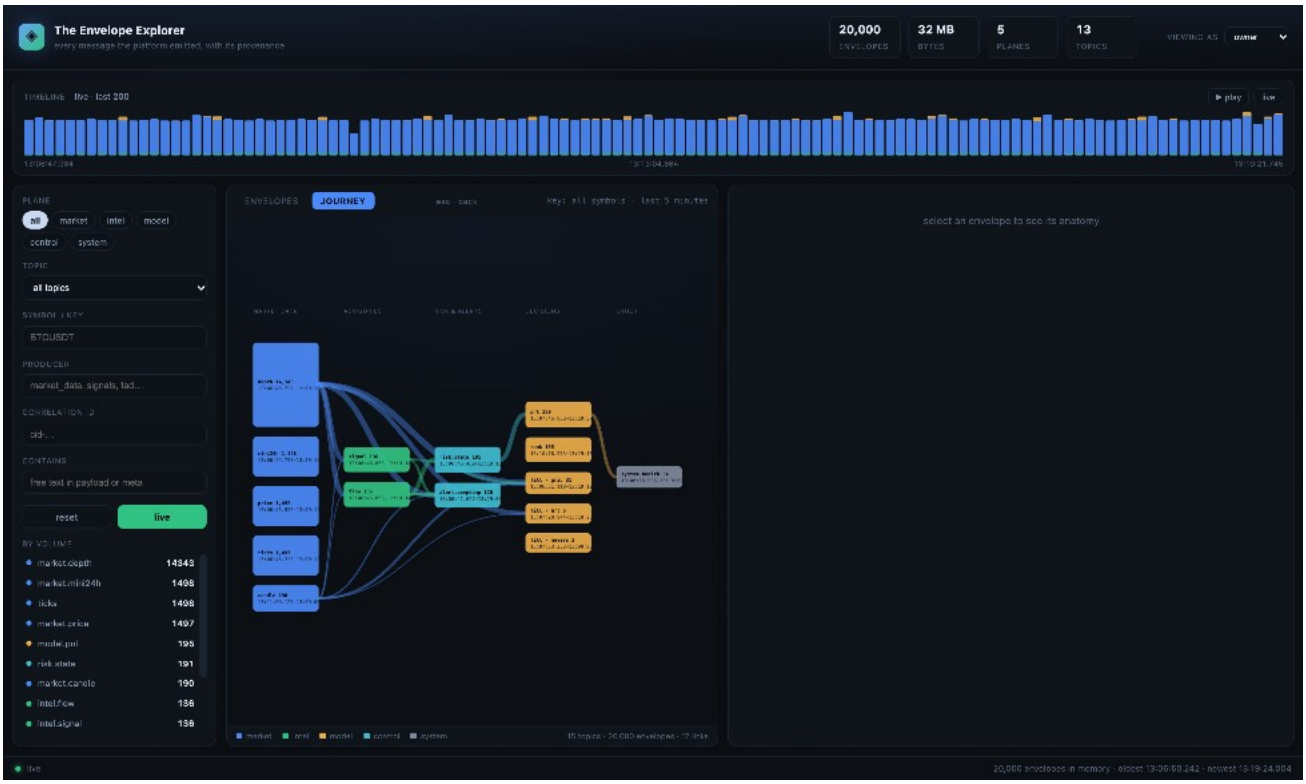


Figure 4 — the whole platform as a flow: five stages and fifteen nodes, including each model that wrote a fill. Box height is volume; ribbon width is the number of correlations observed.

6.1 Two ways to open it

Way	What you get
Select an envelope, then open JOURNEY	the flow opens at that message: two minutes of lead-in, five minutes forward, on its symbol. The node containing it is ringed.
Open JOURNEY with nothing selected	the last five minutes across all symbols — a live view of the platform's shape.

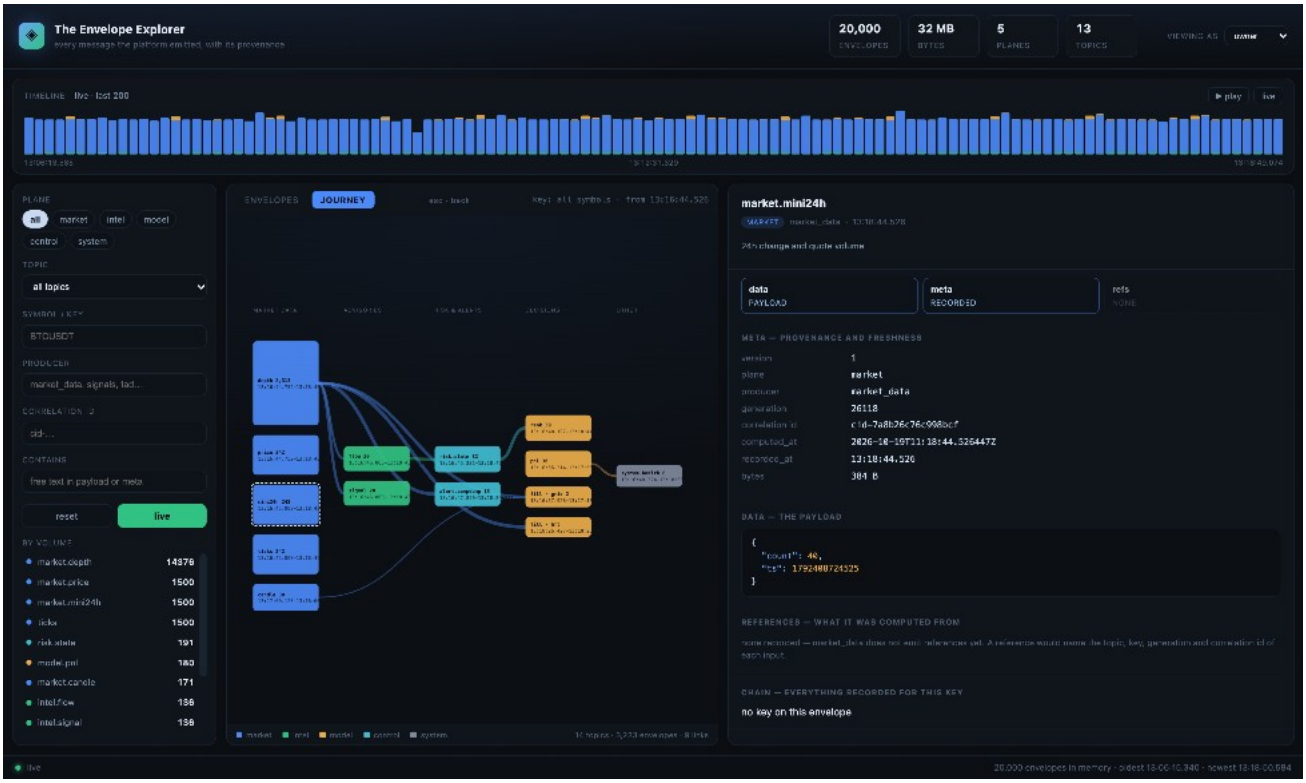


Figure 5 — the same diagram anchored on a model.fill envelope: the dashed ring marks where you came in, and the window runs forward from that moment.

6.2 Reading the diagram

- Columns are the pipeline, left to right: market data → advisories → risk & alerts → decisions → other.
- Box height is volume. market.depth is tall; a handful of fills is a sliver.
- Box label is the topic and its count; the small line beneath is the time span those messages cover.
- Ribbon width is the number of observed correlations between the two topics.
- Colour is the plane: blue market, green intel, amber model, teal control, grey system.
- Nodes are split by producer, so three models writing model.fill appear as three boxes, each labelled with its engine.
- The dashed ring marks the anchor — the message you came in on.

6.3 Moving around

Action	What happens
Click a box	opens the newest envelope of that topic in the right-hand pane
Double-click a box	re-anchors the flow on that node and walks forward from it
Select another envelope in the list	re-anchors the flow there
Esc	back: returns to the previous view and the state you were in — selection, scroll and filters included
Brush the timeline	the flow uses that window too

6.4 A worked walk

1. Find a model.decision row in the list (filter plane=model, topic=model.decision).
2. Open JOURNEY. The ring shows where the decision sits in the flow.
3. Read left: the market data and advisories that preceded it.
4. Double-click the fill box to re-anchor on the fill and see what followed it.

Honest caveat: ribbons are temporal correlations — 'next stage for the same symbol within 120 seconds' — not declared references, because the Python books do not emit refs yet. When they do, the same diagram becomes literal lineage with no

interpretation required.

7 The execution trace: what a model actually used

The journey answers 'how does traffic flow'. The execution trace answers a narrower and more useful question: for this one fill, which raw envelopes did the model read, in what order, and how old were they when it acted?

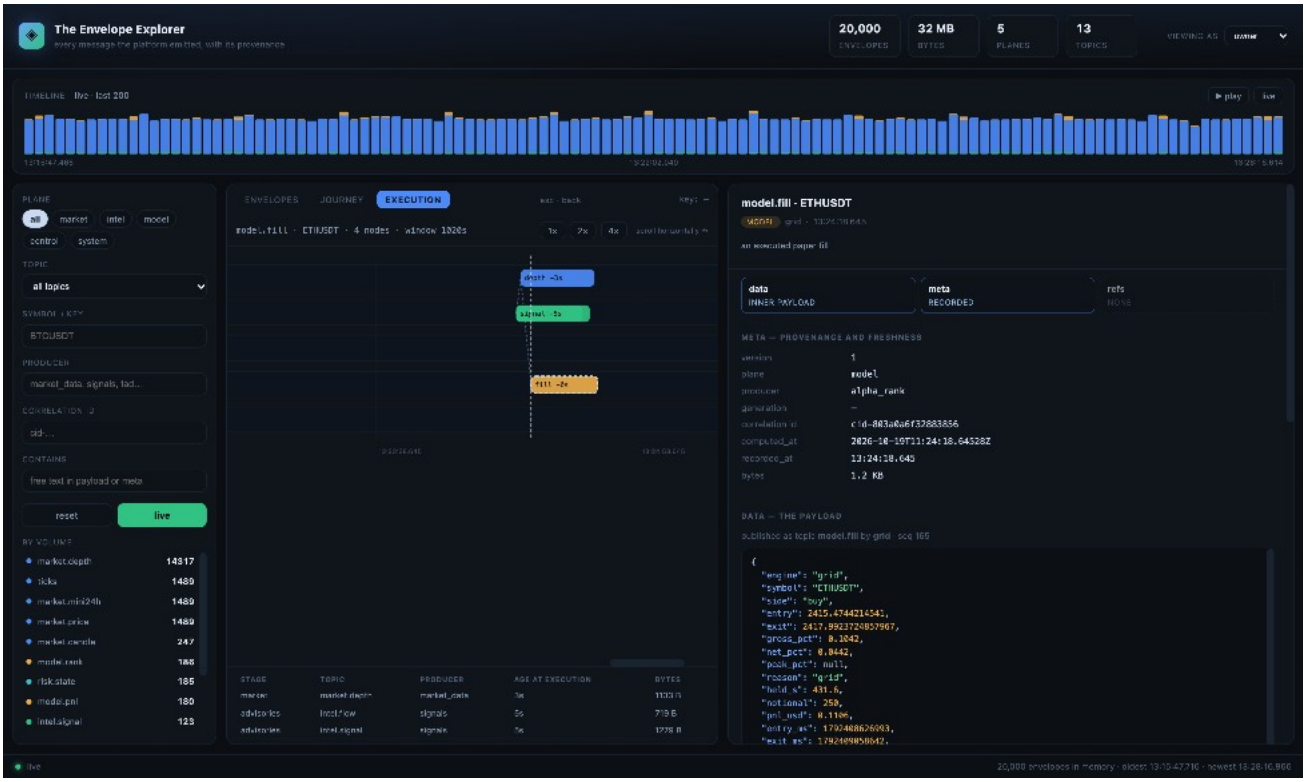


Figure 10 — the execution trace for one model.fill: lanes by stage, nodes placed in time, the dashed line marking the moment of execution, and the inputs listed beneath with the age each had when the model acted.

7.1 How to read it

Element	Meaning
Lanes	one row per stage: market, advisories, risk, model, other
Node	one envelope. The label is the topic and how old it was at execution (−1s means one second before the fill)
Dashed vertical line	the moment of execution — the anchor envelope
Dotted path	the reading order: stage by stage through the chosen envelopes
1× / 2× / 4×	zoom. The canvas scrolls horizontally, so a fifteen-minute window stays readable
Table beneath	the inputs the anchor used: stage, topic, producer, age at execution, bytes

7.2 Working through an execution

1. Open the EXECUTION tab with a fill selected (filter topic = model.fill to find one).
2. The view scrolls itself to the moment of execution and marks it with a dashed line.
3. Read left to right: market data first, then the advisories, then anything that followed the fill.
4. Click any node to see its raw envelope in the right-hand pane — data, meta and references.
5. Use the table beneath to compare ages: a decision made on a 600-second-old candle is a different decision from one made on a 2-second-old candle.

This is the view for 'why did the model do that'. It shows the exact envelopes, their ages, and their producers — not a summary of them. When books start emitting references, the dotted path becomes literal lineage instead of a time-ordered reading of the

window.

8 How and why a model traded — or did not

This is the question a reviewer actually asks. The WHY tab answers it directly: every decision the models recorded, split into the trades they took and the trades they refused, with the rule that stopped them.

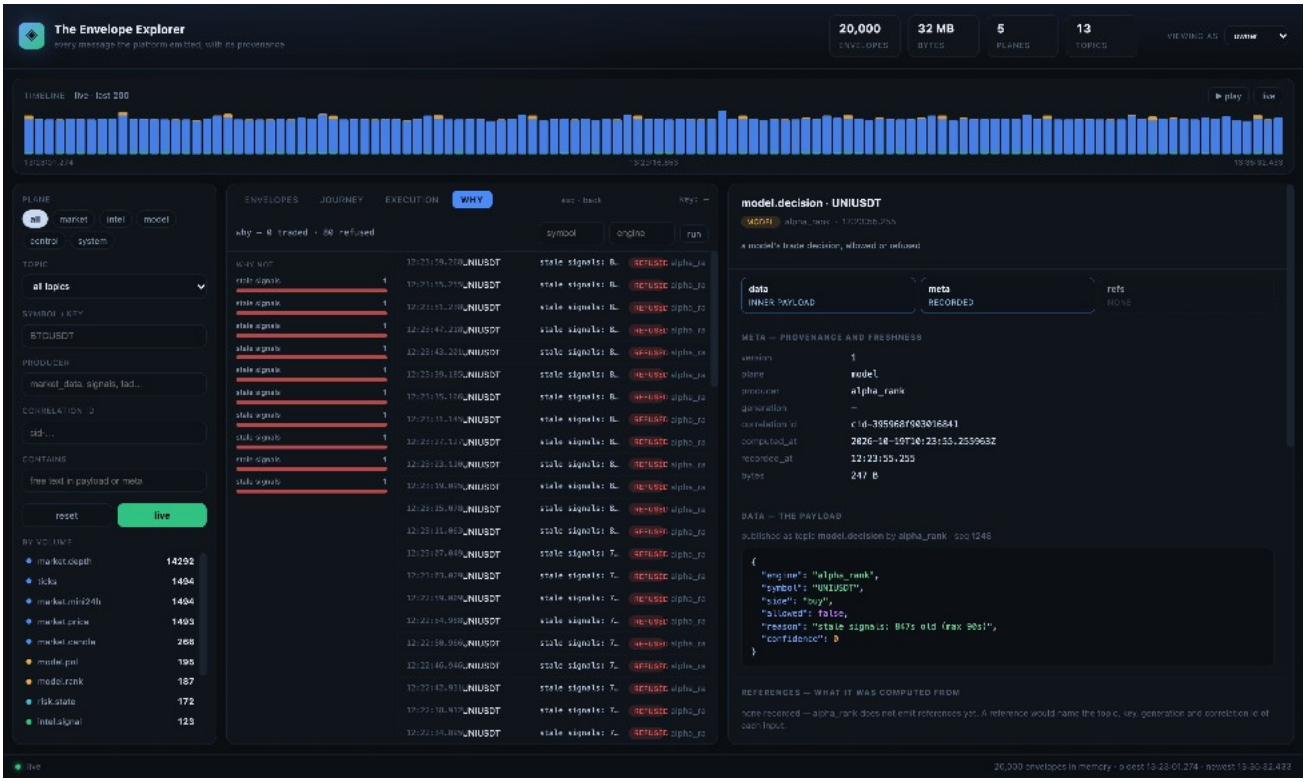


Figure 11 — the WHY tab: reason tally on the left, decisions on the right, and the raw refusal envelope open beside them.

8.1 Reading the tab

Part	What it tells you
Header	how many decisions the scan found: traded, refused, and the symbol if filtered
Why not (left)	the refusal reasons, tallest first. 'stale signals' dominating means the feed is late, not that the model is picky
Decision rows	time, symbol, reason as the model recorded it, outcome, engine
Right pane	the raw decision envelope: data, meta, references — including allowed and reason verbatim
Filters	symbol and engine; Enter runs the scan

8.2 What a decision actually contains

A model.decision payload is short on purpose. It carries the engine, the symbol and side, whether the governor allowed it, the reason, and the confidence the model had. Everything else is in the envelope: who produced it, when, and the correlation id that ties it to the rest of the chain.

Field	Example	Why it matters
engine	alpha_rank	which model made the call
symbol / side	UNIUSDT · buy	what it wanted to do
allowed	false	the governor's verdict
reason	stale signals: 847s old (max 90s)	the rule and the number that tripped it
confidence	0	the model's own conviction — 0 means none

8.3 From refusal to evidence in three clicks

1. Open WHY and find the refusal you care about — the reason is already visible in the row.
2. Click it: the right pane shows the raw envelope, so the reason is quoted from the record, not paraphrased.
3. Double-click it: the EXECUTION tab opens on that decision, with the inputs it read and how old each was.

A refusal is evidence, not noise. 'stale signals: 847s old' names the rule, the age and the limit, and the envelope that says it is the one the platform stored. That is the difference between 'it did not trade' and 'we cannot explain why it did not trade'.

9 Roles and redaction

The 'viewing as' selector in the header changes what you are allowed to see. It is a real filter applied on the server, not a cosmetic setting.

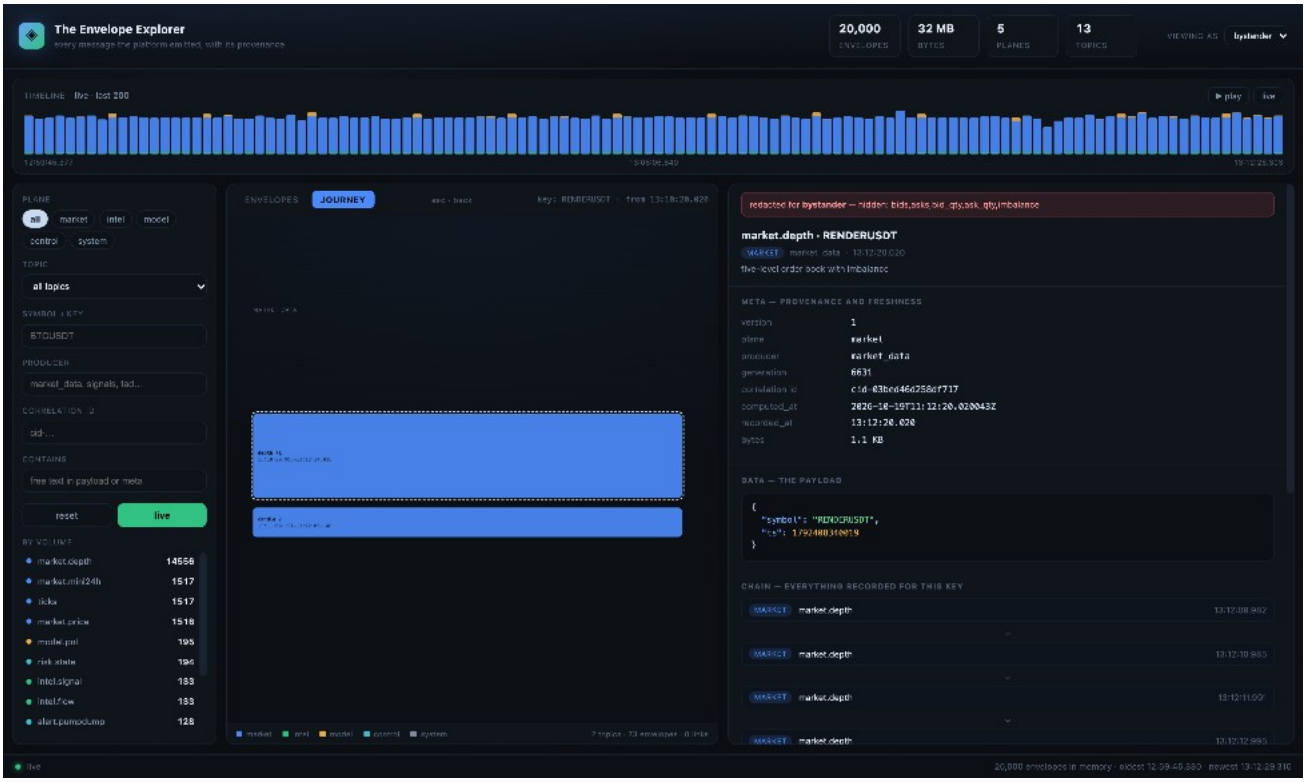


Figure 12 — the same envelope viewed as a bystander: depth fields removed, and a red band naming exactly what was withheld.

Role	Sees	Typical use
owner	everything	you are the operator
auditor	prices and sizes removed, everything else	reviewing decisions without market detail
bystander	the shape and provenance only	showing status to someone outside the desk

Every redaction is recorded in the envelope's meta as redacted_for and redacted_fields, so the removal is part of the record rather than a silent edit. If someone asks what you were shown, the answer is in the message.

10 Three worked examples

10.1 Why did we buy this?

1. Filter symbol = BTCUSDT and plane = model.
2. Open the model.decision row you care about.
3. Switch to JOURNEY: the ring sits on the decision; read left for the advisories and market data.
4. Note the correlation id from the meta pane — quote it in any discussion.

10.2 What happened at a specific minute?

1. Drag the timeline across that minute.
2. The list narrows to it; sort your reading by plane using the left panel.
3. Watch for a gap rather than a quiet period: a gap means no producer, not no market.

10.3 Which model is producing decisions?

1. Filter producer = books (or a specific engine) and topic = model.decision.
2. Compare the generation numbers: a stall shows as a flat generation while others climb.
3. Cross-check the recorder's per-topic counts in /stats to confirm delivery, not just display.

11 Keyboard and quick reference

Key or control	Effect
← (crumb bar)	back one step — the same action as Escape
Esc	one step back: the previous view *and* the envelope selected in it; with no history it clears the selection
Tab (browser)	focus the next filter
drag splitter	resize the filters, the list, the detail pane or the timeline
live / paused	follow new envelopes, or freeze the list
▶ play	step the selected window forward
reset	clear every filter and return to live

11.1 What the colours mean

Colour	Plane	Typical topics
blue	market	ticks, market.price, market.depth, market.candle
green	intel	intel.signal, intel.flow, intel.ta, intel.forecast
amber	model	model.decision, model.fill, model.pnl, model.rank
teal	control	risk.state, exec.quote, alert.pumpdump
grey	system	system.health, review.verdict

12 Troubleshooting

Symptom	Likely cause	What to do
List is empty	nothing matches the filters	press reset
Count frozen at 20,000	feed or recorder stopped	check :8801/stats and :8796/health
A topic never appears	its producer is silent	intel.forecast needs 40 bars before it publishes
Journey shows one column	the symbol only has market data	pick a symbol with decisions (filter plane=model first)
Ribbons are missing	no correlations in the window	widen the window or pick a busier symbol
Chain says 'no key'	the topic is a multi-symbol snapshot	expected for market.mini24h and ticks

The Explorer is read-only by construction: it opens the lake, never the bus, and never writes. If something looks wrong in the platform, the Explorer is the wrong place to fix it — but the right place to prove it.